

and the nodes must exchange information with the entire network, keeping the information in sync while discovering more nodes in the given graph structure. With this approach, each node must produce its colour as output (as shown in Figure 6), effectively overcoming the barrier of distributed systems.

Networking

Over a distributed system, processing and storage resources are sitting apart from one another and only the network connects them. They can communicate with one another and share data via message passing. The communication processes follow a set of protocols to pass messages. Some of the widely used models of communication are:

- Hypertext transfer protocol
- Remote procedure call
- Message queues

Hypertext transfer protocol

(HTTP): HTTP is widely used in web-based computing, where the client and server connect over the distributed network. The client sends a request to the server over a network using HTTP and expects a response. In multi-tier applications, the servers can use HTTP to communicate and pass messages to one another.

Remote procedure call (RPC):

RPC is a form of inter-process communication between different processes running in the distributed system. The idea is to allow the calling of procedures located on different machines. RPC is request/response communication. A local instance of RPC runs on each computing node of the distributed systems, where if system A wants to run a remote procedure that is there on system B, A will ask the local instance of RPC to send the request to system B with a set of parameters, while RPC manages the transmission of parameters and other requests in the message over the network. After

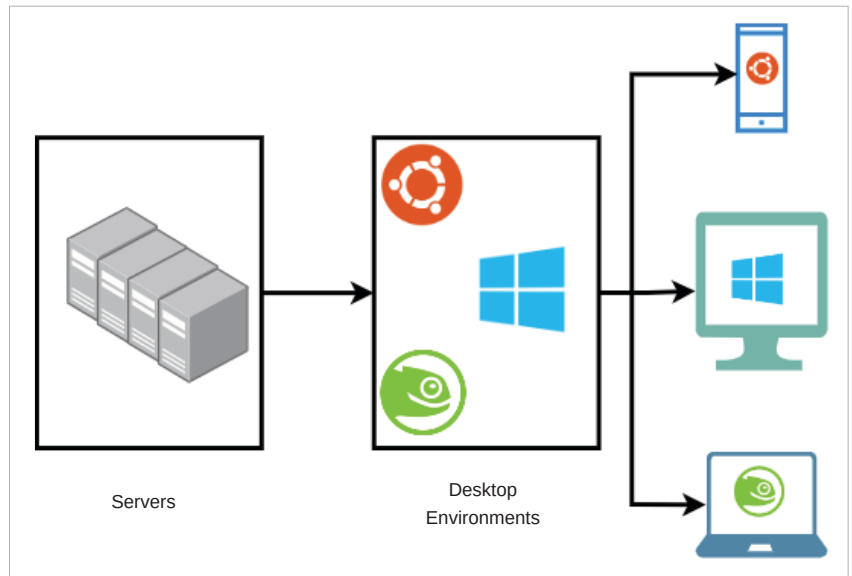


Figure 9: Desktop virtualization

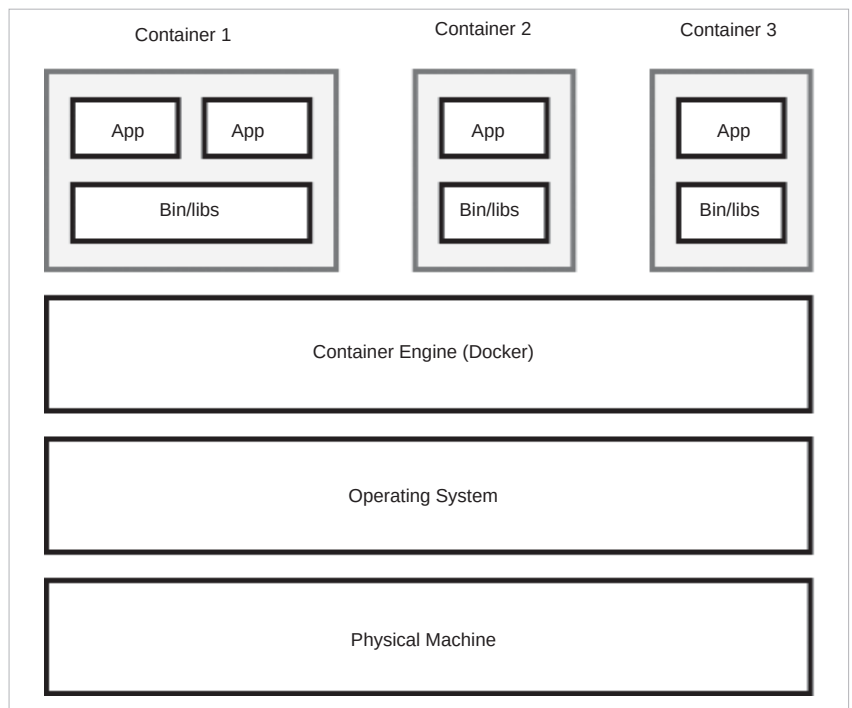


Figure 10: Containerization

the completion of the system, the B procedure is initiated by A, and the information is communicated to the callee from the caller (to A from B).

Message queues: In message queues, we use queues to communicate the message between distributed systems. The message

queue works on the asynchronous communication pattern, where even if the server is down when the message was communicated, it will still be received by the destined server when it comes back up. Each application has its private queue, where messages are delivered. They are stored on the